Bootstrap

Bootstrap

- Twitter Bootstrap is the most popular front end framework in the recent time.
- Bootstrap is an open source toolkit for developing with HTML, CSS, and JS.
- It is sleek, intuitive, and powerful mobile first front-end framework for faster and easier web development.
- It uses HTML, CSS and JavaScript.

Why Use Bootstrap?

- **Mobile first approach** − Bootstrap 3, framework consists of Mobile first styles throughout the entire library instead them of in separate files.
- **Browser Support** − It is supported by all popular browsers.
- **Easy to get started** − With just the knowledge of HTML and CSS anyone can get started with Bootstrap. Also the Bootstrap official site has a good documentation.
- **Responsive design** − Bootstrap's responsive CSS adjusts to Desktops, Tablets and Mobiles.

Why Use Bootstrap?

- Provides a clean and uniform solution for building an interface for developers.
- It contains beautiful and functional built-in components which are easy to customize.
- It also provides web based customization.
- It is an open source.

Bootstrap Applications

- **Scaffolding** − Bootstrap provides a basic structure with Grid System, link styles, and background.
- **CSS** − Bootstrap comes with the feature of global CSS settings, fundamental HTML elements styled and enhanced with extensible classes, and an advanced grid system.
- **Components** − Bootstrap contains over a dozen reusable components built to provide iconography, dropdowns, navigation, alerts, pop-overs, and much more.
- **JavaScript Plugins** − Bootstrap contains over a dozen custom jQuery plugins. You can easily include them all, or one by one.

Using Bootstrap

- You can download the latest version of Bootstrap from https://getbootstrap.com/docs/4.3/getting-started/download/
- Here you will see two versions:
  - Compiled CSS and JS – download the precompiled and minified versions of Bootstrap CSS, JavaScript, and fonts. No documentation or original source code files are included.
  - Source files – compile Bootstrap with your own asset pipeline by downloading our source Sass, JavaScript, and documentation files. This option requires some additional tooling.

Bootstrap CDN

- Skip the download with BootstrapCDN to deliver cached version of Bootstrap's compiled CSS and JS to your project.

```
<link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css"
integrity="sha384-ggOyR0iXCbMQv3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T" crossorigin="anonymous">
<script src=https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/js/bootstrap.min.js
 integrity="sha384-JjSmVgyd0p3pXB1rRibZUAYoIIy6OrQ6VrjIEaFf/nJGzIxFDsf4x0xIM+B07jRM"
crossorigin="anonymous"></script>
```

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File Structure

- Once the compiled version Bootstrap is downloaded, extract the ZIP file, and you will see the following file/directory structure:

```
bootstrap/
├── css/
│   ├── bootstrap.css
│   ├── bootstrap.min.css
│   ├── bootstrap-theme.css
│   └── bootstrap-theme.min.css
├── js/
│   ├── bootstrap.js
│   └── bootstrap.min.js
└── fonts/
    ├── glyphicons-halflings-regular.eot
    ├── glyphicons-halflings-regular.svg
    ├── glyphicons-halflings-regular.ttf
    └── glyphicons-halflings-regular.woff
```

- As you can see, there are compiled CSS and JS (bootstrap.*), as well as compiled and minified CSS and JS (bootstrap.min.*).

- Fonts from Glyphicons are included, as it is the optional Bootstrap theme.

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Source Code File Structure

- If you have downloaded the Bootstrap source code then the file structure would be as follows:

```
├── less/
├── js/
├── fonts/
├── dist/
│   ├── css/
│   ├── js/
│   └── fonts/
├── docs-assets/
├── examples/
└── *.html
```

- The files under *less/*, *js/*, and *fonts/* are the source code for Bootstrap CSS, JS, and icon fonts (respectively).
- The *dist/* folder includes everything listed in the precompiled download section above.
- *docs-assets/*, *examples/*, and all **.html* files are Bootstrap documentation.

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Grid System

- What is a grid?
  - In graphic design, a grid is a structure (usually two-dimensional) made up of a series of intersecting straight (vertical, horizontal) lines used to structure the content.
  - It is widely used to design layout and content structure in print design. In web design, it is a very effective method to create a consistent layout rapidly and effectively using HTML and CSS.

- To put in simple words, grids in web design organise and structure content, makes the websites easy to scan and reduces the cognitive load on users.

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Grid System

- Bootstrap includes a responsive, mobile first fluid grid system that appropriately scales up to 12 columns as the device or viewport size increases.
- It includes predefined classes for easy layout options, as well as powerful mixins for generating more semantic layouts.
- Bootstrap 3 is mobile first in the sense that the code for Bootstrap now starts by targeting smaller screens like mobile devices, tablets, and then “expands” components and grids for larger screens such as laptops, desktops.

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Grid System – Mobile First Strategy

- Content:
  - Determine what is most important.
- Layout:
  - Design to smaller widths first.
  - Base CSS address mobile device first; media queries address for tablet, and desktops.
- Progressive Enhancement:
  - Add elements as screen size increases.

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Working of Bootstrap Grid System

- Grid systems are used for creating page layouts through a series of rows and columns that house your content.

- Here's how the Bootstrap grid system works:
  - Rows must be placed within a **.container** class for proper alignment and padding.

  - Use rows to create horizontal groups of columns.

  - Content should be placed within the columns, and only columns may be the immediate children of rows.

  - Predefined grid classes like **.row and .col-xs-4** are available for quickly making grid layouts.

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Working of Bootstrap Grid System

- Here's how the Bootstrap grid system works:
  - Columns create gutters (gaps between column content) via padding. That padding is offset in rows for the first and the last column via negative margin on **.rows**.
  - Grid columns are created by specifying the number of twelve available columns you wish to span. For example, three equal columns would use three **.col-xs-4**.

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Media Queries

- Media query is a really fancy term for "conditional CSS rule".
- It simply applies some CSS, based on certain conditions set forth. If those conditions are met, the style is applied.
- Media Queries in Bootstrap allow you to move, show and hide content based on the viewport size.

```
/* Extra small devices (phones, less than 768px) */
/* No media query since this is the default in Bootstrap */

/* Small devices (tablets, 768px and up) */
@media (min-width: @screen-sm-min) { ... }

/* Medium devices (desktops, 992px and up) */
@media (min-width: @screen-md-min) { ... }

/* Large devices (large desktops, 1200px and up) */
@media (min-width: @screen-lg-min) { ... }
```

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Media Queries

- Media queries have two parts, a device specification and then a size rule. In the above case, the following rule is set.

- Let us consider this line:

```
@media (min-width: @screen-sm-min) and (max-width: @screen-sm-max) { ... }
```

- For all devices no matter what kind with *min-width: @screen-sm-min* if the width of the screen gets smaller than *@screen-sm-max*, *then do something*.

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Grid Options

	Extra small devices Phones (<768px)	Small devices Tablets (≥768px)	Medium devices Desktops (≥992px)	Large devices Desktops (≥1200px)
Grid behavior	Horizontal at all times	Collapsed to start, horizontal above breakpoints	Collapsed to start, horizontal above breakpoints	Collapsed to start, horizontal above breakpoints
Max container width	None (auto)	750px	970px	1170px
Class prefix	**.col-xs-**	**.col-sm-**	**.col-md-**	**.col-lg-**
# of columns	12	12	12	12
Max column width	Auto	60px	78px	95px
Gutter width	30px (15px on each side of a column)	30px (15px on each side of a column)	30px (15px on each side of a column)	30px (15px on each side of a column)
Nestable	Yes	Yes	Yes	Yes
Offsets	Yes	Yes	Yes	Yes
Column ordering	Yes	Yes	Yes	Yes

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Infrastructure

- Bootstrap makes use of certain HTML elements and CSS properties that require the use of the HTML5 doctype.

- Hence include the below piece of code for HTML5 doctype at the beginning of all your projects using Bootstrap.

```
<!DOCTYPE html>
<html>
<head>
<title>Title of the document</title>
</head>

<body>
The content of the document......
</body>

</html>
```

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Mobile First

- Since Bootstrap 3 has been launched, Bootstrap has become mobile first.
- It means 'mobile first' styles can be found throughout the entire library instead of them in separate files.
- You need to add the **viewport meta tag** to the **<head>** element, to ensure proper rendering and touch zooming on mobile devices.

```
<meta name = "viewport" content = "width = device-width, initial-scale = 1.0">
```

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Mobile First

```
<meta name = "viewport" content = "width = device-width, initial-scale = 1.0">
```

- *width* property controls the width of the device. Setting it to *device-width* will make sure that it is rendered across various devices (mobiles, desktops, tablets...) properly.

- *initial-scale = 1.0* ensures that when loaded, your web page will be rendered at a 1:1 scale, and no zooming will be applied out of the box.

- Add **user-scalable = no** to the **content** attribute to disable zooming capabilities on mobile devices.

- Users are only able to scroll and not zoom with this change, and results in your site feeling a bit more like a native application.

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Responsive Images

- Bootstrap 3 allows you to make the images responsive by adding a class **.img-responsive** to the **<img>** tag.

```
<img src = "..." class = "img-responsive" alt = "Responsive image">
```

- This class applies **max-width: 100%;** and **height: auto;** to the image so that it scales nicely to the parent element.

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Typography and Links

- **Basic Global display** − Sets *background-color: #fff;* on the <*body*> element.
- **Typography** − Uses the *@font-family-base*, *@font-size-base*, and *@line-height-base* attributes as the typographic base.
- **Link styles** − Sets the global link color via attribute *@link-color* and apply link underlines only on *:hover*.
- Bootstrap uses Helvetica Neue, Helvetica, Arial, and sans-serif in its default font stack.

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Forms

- Bootstrap contains HTML markup and extended classes for different styles of forms.
- Bootstrap provides you with following types of form layouts:
  - Vertical (default) form
  - In-line form
  - Horizontal form

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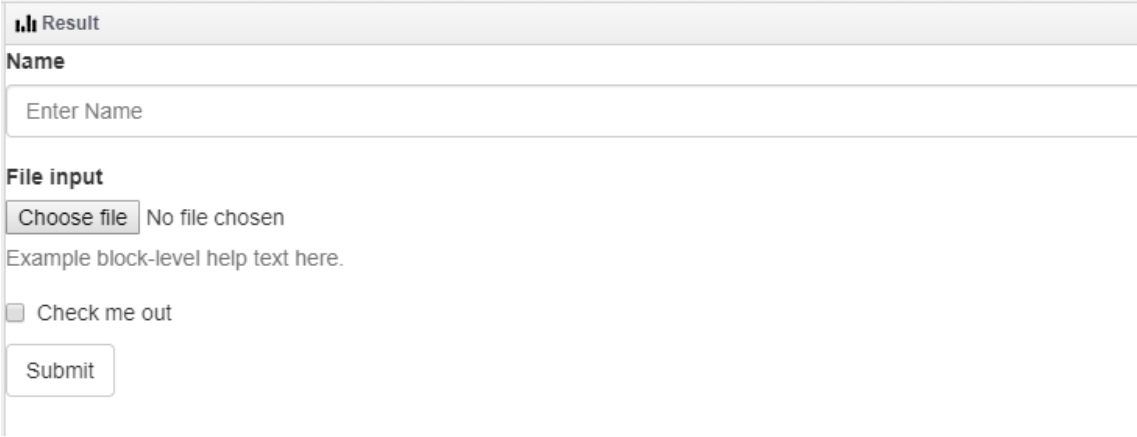
Vertical Form

- The basic form structure comes with Bootstrap; individual form controls automatically receive some global styling.

- To create a basic form:
  - Add a role *form* to the parent <form> element.
  - Wrap labels and controls in a with class *.form-group*. This is needed for optimum spacing.
  - Add a class of *.form-control* to all textual <input>, <textarea>, and <select> elements.



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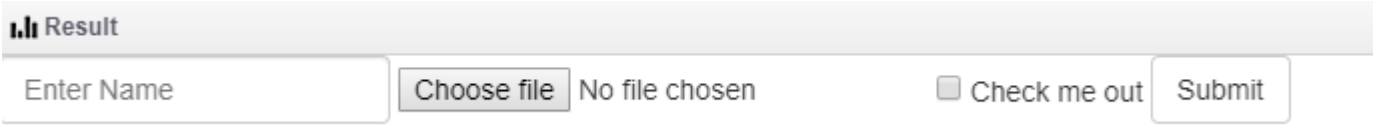
Inline Form

- To create a form where all of the elements are inline, left aligned and labels are alongside, add the class *.form-inline* to the <form> tag.
- By default inputs, selects, and textareas have 100% width in Bootstrap. You need to set a width on the form controls when using inline form.
- Using the class *.sr-only* you can hide the labels of the inline forms.



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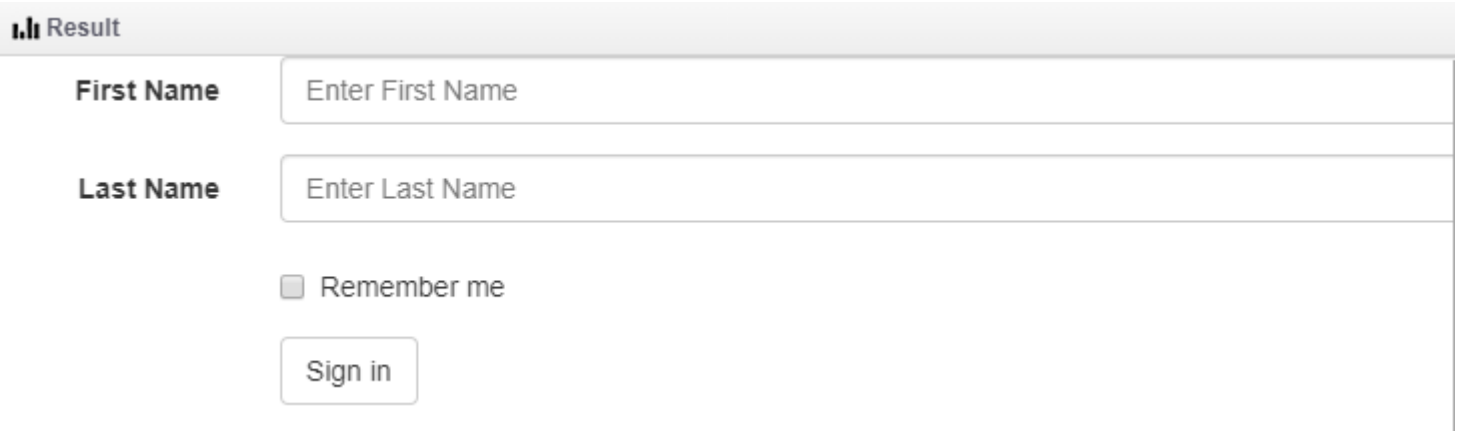
Horizontal Form

- Horizontal forms stands apart from the others not only in the amount of markup, but also in the presentation of the form.

- To create a form that uses the horizontal layout:
  - Add a class of *.form-horizontal* to the parent <form> element.
  - Wrap labels and controls in a with class *.form-group*.
  - Add a class of *.control-label* to the labels.



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Inputs

- Bootstrap natively supports the most common form controls mainly *input, textarea, checkbox, radio, and select.*
- The most common form text field is the input field. This is where users will enter most of the essential form data.
- Bootstrap offers support for all native HTML5 input types.
- The textarea is used when you need multiple lines of input. Change *rows* attribute as necessary (fewer rows = smaller box, more rows = bigger box).

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Inputs

- When building a form, use *checkbox* if you want the user to select any number of options from a list. Use *radio* if you want to limit the user to just one selection.
- Use *.checkbox-inline* or *.radio-inline* class to a series of checkboxes or radios for controls appear on the same line.
- Use <select> for list options with which the user is familiar, such as states or numbers.
- Use *multiple = "multiple"* to allow the users to select more than one option.

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Focus

- In addition to the *:focus* (i.e., a user clicks into the input or tabs onto it) state, Bootstrap offers styling for disabled inputs and classes for form validation.

- When an input receives *:focus*, the outline of the input is removed and a *box-shadow* is applied.

- If you need to disable an input, simply adding the *disabled* attribute will not only disable it; it will also change the styling and the mouse cursor when the cursor hovers over the element.

- Add the disabled attribute to a <fieldset> to disable all the controls within the <fieldset> at once.

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Buttons

- Anything that is given a class of **.btn** will inherit the default look of a gray button with rounded corners. However, Bootstrap provides some options to style buttons

Sr.No.	Class & Description
1	**btn** Default/ Standard button.
2	**btn-primary** Provides extra visual weight and identifies the primary action in a set of buttons.
3	**btn-success** Indicates a successful or positive action.
4	**btn-info** Contextual button for informational alert messages.
5	**btn-warning** Indicates caution should be taken with this action.
6	**btn-danger** Indicates a dangerous or potentially negative action.
7	**btn-link** Deemphasize a button by making it look like a link while maintaining button behavior.

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Buttons

- The following table summarizes the classes used to get buttons of various sizes:

Sr.No.	Class & Description
1	**.btn-lg** This makes the button size large.
2	**.btn-sm** This makes the button size small.
3	**.btn-xs** This makes the button size extra small.
4	**.btn-block** This creates block level buttons—those that span the full width of a parent.

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Button State

- Bootstrap provides classes which allow you to change the state of buttons as active, disabled etc.
- Active State – Buttons will appear pressed (with a darker background, darker border, and inset shadow) when active.
- Disabled State – When you disable a button, it will fade in colour by 50%, and lose the gradient.
- You may use button classes with <a>, <button>, or <input> element.
- But it is recommended that you use it with <button> elements mostly to avoid cross browser inconsistency issues.

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Images

- Bootstrap provides three classes that can be used to apply some simple styles to images:
- ***.img-rounded*** − adds *border-radius:6px* to give the image rounded corners.
- ***.img-circle*** − makes the entire image round by adding *border-radius:500px*.
- ***.img-thumbnail*** − adds a bit of padding and a gray border.

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Helper Classes

- Use the generic close icon for dismissing content like modals and alerts. Use the class **close** to get the close icon.
- Use carets to indicate dropdown functionality and direction. To get this functionality use the class **caret** with a element.
- You can float an element to the left or right with class **pull-left** or **pull-right** respectively.
- Use class **center-block** to set an element to center.
- You can hide an element to all devices except screen readers with the class **.sr-only**.

Questions